Índice general

Capítulo 1

Formulas Used in This Topic

1.0.1. Expected Value of a Discrete Random Variable

$$E[X] = \sum_{i=1}^{n} x_i p_i$$

Where

- X random variable representing uncertain outcomes
- x_i the i-th possible outcome of X
- p_i probability associated with outcome x_i
- n number of possible outcomes

Context Represents the long-run average outcome under repeated identical conditions.

1.0.2. Variance of a Discrete Random Variable

$$\mathrm{Var}(X) = \sum_{i=1}^{n} (x_i - E[X])^2 p_i$$

Where

- x_i possible outcome
- $E[X]$ expected value of the distribution
- p_i probability of outcome x_i
- n number of outcomes

Context Measures dispersion and quantifies risk or volatility.

1.0.3. Expected Monetary Value (EMV)

$$EMV(A_j) = \sum_{i=1}^{m} p_i \, \text{Payoff}(A_j, S_i)$$

Where

- A_j decision alternative (for example a strategy)
- S_i state of nature (future scenario)
- $\text{Payoff}(A_j, S_i)$ monetary result of A_j under state S_i
- p_i probability of state S_i
- m number of states of nature

Context Represents the average monetary performance of a strategy under uncertainty.

1.0.4. Maximin Criterion

$$\text{Maximin}(A_j) = \max_j \left(\min_i \text{Payoff}(A_j, S_i) \right)$$

Where

- A_j decision alternatives
- S_i states of nature
- $\min_i \text{Payoff}(A_j, S_i)$ worst-case payoff of A_j
- outer max selects the highest worst-case result

Context Represents a conservative decision maker seeking safety.

1.0.5. Maximax Criterion

$$\text{Maximax}(A_j) = \max_j \left(\max_i \text{Payoff}(A_j, S_i) \right)$$

Where

- inner max best payoff each alternative can achieve
- outer max selects the alternative with the largest best-case payoff

Context Represents an optimistic decision maker seeking maximum possible gain.

Back to Top

Capítulo 2

Example 1

2.0.1. Problem description

A technology firm is planning its strategy for the next year and must choose exactly one of three possible options. One option is to innovate (Strategy A) by launching a new, cutting-edge product. Another option is to improve the current product line (Strategy B) by upgrading and refining what the firm already sells. The third option is to license an external technology (Strategy C) instead of developing new technology in-house.

The firm knows that future market demand is uncertain and can be in one of three states high demand, moderate demand or low demand. Based on market research, management estimates that the probability of high demand is $0{,}30$, the probability of moderate demand is $0{,}40$ and the probability of low demand is $0{,}30$. These probabilities are treated as the best available information before any decision is made.

For each combination of strategy and demand, the firm has an estimate of the profit measured in thousands of dollars. If demand is high, the expected profits are 150 for Innovate (A), 110 for Improve (B) and 70 for License (C). If demand is moderate, the expected profits are 80 for Innovate (A), 90 for Improve (B) and 70 for License (C). If demand is low, the expected profits are -20 for Innovate (A), 40 for Improve (B) and 60 for License (C). The firm wants to analyse this decision using several decision-making rules, including expected value, maximax, maximin and the maximum opportunity minimax regret criterion, and then determine which strategy provides the most coherent choice given its view of risk and the available probability information.

2.0.2. Solution

(C2) Interpreting the parts of the decision problem

We identify the key elements and summarise them in a table.

Element	Description
Decision alternatives	Innovate (A), Improve (B), License (C)
States of nature	High demand, Moderate demand, Low demand
Fortuitous events	The actual market demand that will occur, which is unknown when deciding
Consequences	Profit in thousands of dollars for each strategy–demand combination
Probabilities	High demand $0{,}30$, Moderate demand $0{,}40$, Low demand $0{,}30$

(C3) Building the payoff table from the description

From the problem description, we construct the payoff table (profits in thousands of dollars).

State of Nature	Probability	Innovate (A)	Improve (B)	License (C)
High demand	$0{,}30$	150	110	70
Moderate demand	$0{,}40$	80	90	70
Low demand	$0{,}30$	-20	40	60

(C4) Maximax rule decision making without probabilities

We ignore probabilities and focus only on the best possible payoff for each strategy.

Best payoff for each strategy

$$\max_i \text{Payoff}(A, S_i) = 150k$$
$$\max_i \text{Payoff}(B, S_i) = 110k$$
$$\max_i \text{Payoff}(C, S_i) = 70k$$

The largest of these is $150k$, so the Maximax rule recommends *Innovate (A)*.

(C5) Maximin rule decision making without probabilities

Now we ignore probabilities and focus on the worst-case payoff of each strategy.

Worst payoff for each strategy

$$\min_i \text{Payoff}(A, S_i) = -20k$$
$$\min_i \text{Payoff}(B, S_i) = 40k$$
$$\min_i \text{Payoff}(C, S_i) = 60k$$

We then pick the strategy with the highest of these worst outcomes

$$\max\{-20, 40, 60\} = 60k$$

The Maximin rule recommends *License (C)*.

(C6) Maximum opportunity criterion minimax regret

We construct the regret (opportunity loss) table. For each state of nature, we compare each payoff with the best payoff in that state.

Best payoff in each state

$$\begin{aligned} \text{High demand}: &\ \max\{150, 110, 70\} = 150k \\ \text{Moderate demand}: &\ \max\{80, 90, 70\} = 90k \\ \text{Low demand}: &\ \max\{-20, 40, 60\} = 60k \end{aligned}$$

Regrets are computed as best payoff minus actual payoff.

State of Nature	Innovate (A)	Improve (B)	License (C)
High demand	$150 - 150 = 0$	$150 - 110 = 40$	$150 - 70 = 80$
Moderate demand	$90 - 80 = 10$	$90 - 90 = 0$	$90 - 70 = 20$
Low demand	$60 - (-20) = 80$	$60 - 40 = 20$	$60 - 60 = 0$

Maximum regret for each strategy

$$\max_i \text{Regret}(A, S_i) = \max\{0, 10, 80\} = 80$$
$$\max_i \text{Regret}(B, S_i) = \max\{40, 0, 20\} = 40$$
$$\max_i \text{Regret}(C, S_i) = \max\{80, 20, 0\} = 80$$

The minimax regret rule chooses the strategy with the smallest maximum regret, so the maximum opportunity criterion recommends *Improve (B)*.

(C7) Using probabilities and expected value Bayes decision rule

We apply the Expected Monetary Value (EMV) rule. For each strategy, we multiply each payoff by the probability of its state and add the results.

For Innovate (A)

$$\begin{aligned} EMV_A &= 0{,}30(150) + 0{,}40(80) + 0{,}30(-20) \\ &= 0{,}30 \cdot 150 + 0{,}40 \cdot 80 + 0{,}30 \cdot (-20) \\ &= 45 + 32 - 6 = 71k \end{aligned}$$

For Improve (B)

$$\begin{aligned} EMV_B &= 0{,}30(110) + 0{,}40(90) + 0{,}30(40) \\ &= 0{,}30 \cdot 110 + 0{,}40 \cdot 90 + 0{,}30 \cdot 40 \\ &= 33 + 36 + 12 = 81k \end{aligned}$$

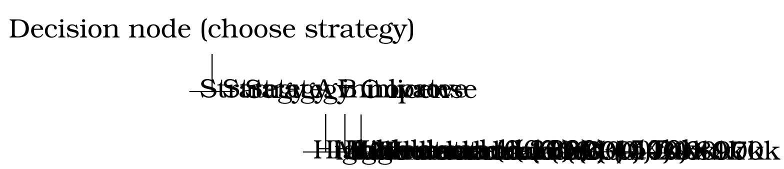
For License (C)

$$\begin{aligned} EMV_C &= 0{,}30(70) + 0{,}40(70) + 0{,}30(60) \\ &= 0{,}30 \cdot 70 + 0{,}40 \cdot 70 + 0{,}30 \cdot 60 \\ &= 21 + 28 + 18 = 67k \end{aligned}$$

Under the Bayes EMV rule, the best strategy is the one with the highest EMV, so EMV recommends *Improve (B)* with $81k$.

(C9) Tree diagram of the decision-making problem

We represent the same situation as a decision tree.



This diagram helps visualise all possible paths decisions followed by demand realisations and the associated payoffs.

(C8) Comparing Bayes, Maximin, Maximax and minimax regret

Summary of recommendations from the different rules

- Bayes EMV expected value with probabilities, no experimentation recommends *Improve (B)*.

- Maximum opportunity minimax regret recommends *Improve (B)*.
- Maximax very optimistic, ignores probabilities recommends *Innovate (A)*.
- Maximin very cautious, ignores probabilities recommends *License (C)*.

Bayes and minimax regret use the full payoff structure (and, in the case of Bayes, the probabilities) to favor a more balanced option, whereas Maximax and Maximin focus only on the most extreme possible outcomes.

(C10) Evaluating all possibilities in the decision problem

Considering all the strategies and rules together, the firm can reflect on the consequences of each possible choice. Innovate (A) offers the highest possible payoff but also the lowest worst-case payoff and a high maximum regret. License (C) protects strongly against the worst case but sacrifices expected profit and is not favoured by EMV or regret. Improve (B) is the only strategy that is simultaneously supported by expected value and by the minimax regret rule. This evaluation shows how different attitudes toward risk and regret lead to different recommended actions, and highlights Improve (B) as a compromise between risk and return.

(C1) Formulating the final decision and summarising all criteria

Based on each of the worked decision criteria, the recommended decision would be as follows.

If the firm decides according to the Maximax rule (C4) and is willing to act with a very optimistic, risk-seeking attitude, it should choose *Innovate (A)* because this strategy has the highest possible payoff among all options.

If the firm decides according to the Maximin rule (C5) and adopts a very conservative, worst-case oriented perspective, it should choose *License (C)* since this strategy has the strongest worst-case payoff.

If the firm decides according to the maximum opportunity minimax regret rule (C6) and wants to minimise the largest possible regret, it should choose *Improve (B)* because this strategy has the smallest maximum regret.

If the firm decides according to the Bayes EMV rule (C7) using the estimated probabilities and focusing on long-run average profit, it should again choose *Improve (B)* because it has the highest expected monetary value.

The decision tree (C9) does not change the recommendation but confirms, in a visual and systematic way, the same payoffs and probabilities used in these rules.

Finally, the comparison in (C8) and the global evaluation in (C10) show that each rule corresponds to a particular attitude toward risk and uncertainty.

Taking all criteria together, a coherent summary conclusion is that the strategy *Improve (B)* provides the most balanced decision for this firm. It is favoured by the expected value approach and by the minimax regret criterion, it avoids the very poor worst case of Innovate (A) and it still offers better long-run performance than License (C). Therefore, in the context of this problem and given the available information, the firm should formulate its final decision as adopting the strategy *Improve (B)* for the next year.

Back to Top

Capítulo 3

Example 2

3.0.1. Problem description

A Colombian agro-export firm that currently sells fresh avocados and coffee in several markets is planning a long-term investment strategy. Management is considering three mutually exclusive options for the next five years. The first possibility is to build a modern avocado processing plant focused on the European Union market, which will be referred to as Strategy A and described as the EU avocado plant. The second possibility is to concentrate on raw coffee exports to the United States, improving logistics and marketing for that specific product, which will be referred to as Strategy B or US coffee focus. The third possibility is to diversify its exports across several regional markets in Latin America and to maintain a smaller presence in both the European Union and the United States, which will be referred to as Strategy C or regional diversification.

The firm is worried about uncertainty in two independent aspects that affect its profit. The first aspect is global demand for its export products, which affects revenue. The second aspect is production and logistics costs, which affect expenses. Management models global demand with three possible demand scenarios. When world demand is high, the EU avocado plant is expected to generate annual revenue of 7000, the US coffee focus 5000 and regional diversification 4800. When demand is moderate, the EU avocado plant is expected to generate revenue of 4800, the US coffee focus 4200 and regional diversification 4300. When demand is weak, the EU avocado plant is expected to generate revenue of 3200, the US coffee focus 3600 and regional diversification 3900. These revenue figures are measured in thousands of US dollars per year. The firm estimates that the probability of high demand is 0.30, the probability of moderate demand is 0.45 and the probability of weak demand is 0.25.

Independent of demand, the firm also considers two possible cost scenarios for production and logistics. In a favourable cost scenario with low input and shipping costs, yearly expenses are expected to be 2300 for the EU avocado plant, 1600 for the US coffee focus and 2000 for regional diversification. In an unfavourable cost scenario with high fuel prices and shipping costs, yearly expenses are expected to be 3500 for the EU avocado plant, 2600 for the US coffee focus and 2700 for regional diversification. These expense figures are also measured in thousands of US dollars per year. The firm estimates that the probability of favourable costs is 0.60 and the probability of unfavourable costs is 0.40, and it assumes that demand conditions and cost conditions are independent.

Profit in each year is defined as revenue minus expenses. Therefore, profit for each strategy in any given year will be determined by the combination of the demand scenario that occurs and the cost scenario that occurs. Because the firm considers the demand event and the cost event to be independent, the probability of any combined scenario is the product of the probability of the corresponding demand level and the probability of the corresponding cost level. Management wants to analyse these three strategies using different decision rules, including Maximax, Maximin, the maximum opportunity minimax regret criterion and the Bayes expected value rule, and then decide which strategy is most coherent given its views about risk and its probability assessments for demand and costs.

3.0.2. Solution

(C2) Interpreting the parts of the decision problem

We interpret the narrative and identify the key components of the decision situation.

(C3) Building the payoff table from revenue and expense information

From the problem description, we first organise the revenue and expense information for each separate event. All figures are in thousands of US dollars per year.

Revenue as a function of demand for each strategy:

Expenses as a function of cost conditions for each strategy:

Because demand and cost conditions are independent, a state of nature is described by a pair of events, for example high demand with favourable costs. The probability of each combined state is the product of the probabilities of the corresponding demand and cost levels. Profit is revenue minus expenses for the chosen strategy in that combined state.

We define six combined states of nature: S1 is high demand with favourable costs, S2 is high demand with unfavourable costs, S3 is moderate demand with favourable costs, S4 is moderate demand with unfavourable costs, S5 is weak demand with favourable costs

Element	Description
Decision alternatives	Strategy A is the EU avocado plant, which focuses on processed avocados for the European Union. Strategy B is the US coffee focus, which concentrates on raw coffee exports to the United States. Strategy C is regional diversification, which mixes exports across Latin America and maintains a smaller EU and US presence.
Independent events related to uncertainty	Global demand conditions, which determine revenue, with three levels high, moderate and weak. Cost conditions, which determine expenses, with two levels favourable costs and unfavourable costs.
States of nature	Each state of nature is a combination of one demand level and one cost level, for example high demand with favourable costs, moderate demand with unfavourable costs and so on.
Fortuitous events	The actual demand level and the actual cost level that occur in a given year. These outcomes are outside the firm's control and are unknown when the strategy is chosen.
Consequences	Annual profit measured in thousands of US dollars for each strategy, obtained by subtracting the expense level that corresponds to the cost scenario from the revenue level that corresponds to the demand scenario.
Probabilities	Demand probabilities are 0.30 for high demand, 0.45 for moderate demand and 0.25 for weak demand. Cost probabilities are 0.60 for favourable costs and 0.40 for unfavourable costs. The events demand level and cost level are assumed independent, so the probability of any combined state of nature is the product of the corresponding demand probability and cost probability.

Demand level	Probability	Strategy A Revenue	Strategy B Revenue	Strategy C Revenue
High demand	0.30	7000	5000	4800
Moderate demand	0.45	4800	4200	4300
Weak demand	0.25	3200	3600	3900

Cost level	Probability	Strategy A Expenses	Strategy B Expenses	Strategy C Expenses
Favourable costs	0.60	2300	1600	2000
Unfavourable costs	0.40	3500	2600	2700

and S6 is weak demand with unfavourable costs. The joint probabilities are

$$P(S1) = 0{,}30 \cdot 0{,}60 = 0{,}18, \quad P(S2) = 0{,}30 \cdot 0{,}40 = 0{,}12, \quad P(S3) = 0{,}45 \cdot 0{,}60 = 0{,}27,$$

$$P(S4) = 0{,}45 \cdot 0{,}40 = 0{,}18, \quad P(S5) = 0{,}25 \cdot 0{,}60 = 0{,}15, \quad P(S6) = 0{,}25 \cdot 0{,}40 = 0{,}10.$$

Profit for each strategy and state of nature is revenue minus expenses. For example, under S3, which is moderate demand and favourable costs, the EU avocado plant has revenue 4800 and expenses 2300, so profit is $4800 - 2300 = 2500$. Under S6, which is weak demand and unfavourable costs, the US coffee focus has revenue 3600 and expenses 2600, so profit is $3600 - 2600 = 1000$.

The payoff table of profit in thousands of US dollars for each combined state is:

(C4) Maximax rule for decision making without probabilities

Under the Maximax rule, the decision maker focuses on the best possible payoff of each strategy and ignores both probabilities and worst cases. We identify, for each strategy, the highest profit among the six states.

$$\text{máx}\{4700,\ 3500,\ 2500,\ 1300,\ 900,\ -300\} = 4700.$$

$$\text{máx}\{3400,\ 2400,\ 2600,\ 1600,\ 2000,\ 1000\} = 3400.$$

$$\text{máx}\{2800,\ 2100,\ 2300,\ 1600,\ 1900,\ 1200\} = 2800.$$

The largest of these best profits is 4700, which corresponds to strategy A in state S1. According to the Maximax rule, the firm should choose *Strategy A (EU avocado plant).*

(C5) Maximin rule for decision making without probabilities

Under the Maximin rule, the decision maker looks at the worst profit of each strategy and then chooses the strategy whose worst-case outcome is the strongest.

$$\text{mín}\{4700,\ 3500,\ 2500,\ 1300,\ 900,\ -300\} = -300.$$

$$\text{mín}\{3400,\ 2400,\ 2600,\ 1600,\ 2000,\ 1000\} = 1000.$$

$$\text{mín}\{2800,\ 2100,\ 2300,\ 1600,\ 1900,\ 1200\} = 1200.$$

We compare these worst-case values and choose the highest one,

$$\text{máx}\{-300,\ 1000,\ 1200\} = 1200.$$

Under the Maximin rule, the firm should choose *Strategy C (regional diversification)*, because it offers the strongest worst-case profit.

(C6) Maximum opportunity criterion (minimax regret)

The maximum opportunity or minimax regret criterion focuses on regret, defined as the loss from not having chosen the best strategy in a given state of nature. For each state, we identify the best profit and then compute regret as the best profit in that state minus the profit of the chosen strategy.

$$\begin{aligned}
\text{S1 (high demand, favourable costs)} &: \ \text{máx}\{4700,\ 3400,\ 2800\} = 4700,\\
\text{S2 (high demand, unfavourable costs)} &: \ \text{máx}\{3500,\ 2400,\ 2100\} = 3500,\\
\text{S3 (moderate demand, favourable costs)} &: \ \text{máx}\{2500,\ 2600,\ 2300\} = 2600,\\
\text{S4 (moderate demand, unfavourable costs)} &: \ \text{máx}\{1300,\ 1600,\ 1600\} = 1600,\\
\text{S5 (weak demand, favourable costs)} &: \ \text{máx}\{900,\ 2000,\ 1900\} = 2000,\\
\text{S6 (weak demand, unfavourable costs)} &: \ \text{máx}\{-300,\ 1000,\ 1200\} = 1200.
\end{aligned}$$

The regret table in thousands of US dollars is:

For each strategy, we identify its maximum regret across all states:

$$\begin{aligned}
\text{máx}\,\text{Regret for A} &= \text{máx}\{0,\ 0,\ 100,\ 300,\ 1100,\ 1500\} = 1500,\\
\text{máx}\,\text{Regret for B} &= \text{máx}\{1300,\ 1100,\ 0,\ 0,\ 0,\ 200\} = 1300,\\
\text{máx}\,\text{Regret for C} &= \text{máx}\{1900,\ 1400,\ 300,\ 0,\ 100,\ 0\} = 1900.
\end{aligned}$$

The minimax regret rule chooses the strategy with the smallest of these maximum regrets, which is strategy B. The maximum opportunity criterion therefore recommends *Strategy B (US coffee focus)*.

(C7) Using probabilities and expected value (Bayes decision rule)

The Bayes decision rule with expected monetary value (EMV) multiplies each profit by the probability of its state and adds the results. We use the joint probabilities of S1 to S6.

$$\begin{aligned}
EMV_A &= 0{,}18(4700) + 0{,}12(3500) + 0{,}27(2500) + 0{,}18(1300) + 0{,}15(900) + 0{,}10(-300)\\
&= 846 + 420 + 675 + 234 + 135 - 30 = 2280.
\end{aligned}$$

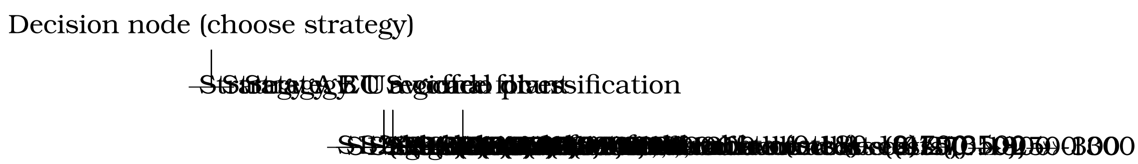
$$EMV_B = 0{,}18(3400) + 0{,}12(2400) + 0{,}27(2600) + 0{,}18(1600) + 0{,}15(2000) + 0{,}10(1000)$$
$$= 612 + 288 + 702 + 288 + 300 + 100 = 2290.$$

$$EMV_C = 0{,}18(2800) + 0{,}12(2100) + 0{,}27(2300) + 0{,}18(1600) + 0{,}15(1900) + 0{,}10(1200)$$
$$= 504 + 252 + 621 + 288 + 285 + 120 = 2070.$$

These values are expected annual profits in thousands of US dollars. Since $EMV_B = 2290$ is slightly higher than $EMV_A = 2280$ and larger than $EMV_C = 2070$, the Bayes EMV rule recommends *Strategy B (US coffee focus)*.

(C9) Tree diagram of the decision-making problem

The same decision situation can be represented as a decision tree. The firm first chooses a strategy, then the demand and cost outcomes are realised. Because demand and costs are independent, their probabilities multiply along each path. For simplicity, we show the combined states with their joint probabilities and profits.



This diagram makes explicit the sequence of decisions and uncertain events, as well as the joint probabilities and profits associated with each possible path.

(C8) Comparing Bayes, Maximin, Maximax and minimax regret

We summarise the recommendations from the different decision rules:

- Maximax, which is very optimistic and focuses on the best possible profit, recommends Strategy A because it can reach 4700.
- Maximin, which is very cautious and focuses on the worst-case profit, recommends Strategy C because its worst case is 1200.
- The maximum opportunity or minimax regret rule, which focuses on limiting the largest possible regret, recommends Strategy B because its maximum regret is 1300, smaller than the maximum regrets of A and C.
- The Bayes EMV rule, which uses the full joint distribution of probabilities and profits, recommends Strategy B because it has the highest expected monetary value.

This comparison shows that rules ignoring probabilities and focusing on extreme outcomes can produce different recommendations from rules that use the full information about the distribution of profits. Strategy A is favoured by a very optimistic view, Strategy C by a very cautious view, and Strategy B by both minimax regret and expected value.

(C10) Evaluating all possibilities in the decision problem

Looking across all rules, the firm can evaluate each strategy in terms of risk and reward. Strategy A, the EU avocado plant, offers the highest possible profit if demand is high and costs are favourable but exposes the firm to a negative profit of -300 when demand is weak and costs are unfavourable. It also has a relatively large maximum regret, because in some unfavourable states another strategy would have generated much better results. Strategy C, regional diversification, never performs extremely badly or extremely well. It offers the strongest worst-case profit but does not maximise expected profit or minimise regret. Strategy B, the US coffee focus, does not reach the highest possible profit of Strategy A but avoids losses, maintains good results in all combined states and is simultaneously favoured by expected value and by the minimax regret criterion. This evaluation highlights how attitudes towards risk, regret and the use of probabilities lead to different recommended actions, and why Strategy B appears as a compromise between the more aggressive and the more conservative options.

(C1) Formulating the final decision and summarising all criteria

Based on each of the decision criteria used in the solution, the recommended strategy would be as follows. If the firm decides according to the Maximax rule and adopts a highly optimistic and risk-seeking attitude, it should choose Strategy A, the EU avocado plant, because this option can reach the highest profit among all combined states. If the firm decides according to the Maximin rule and wants to protect itself against the worst possible profit, it should choose Strategy C, regional diversification, because this option has the strongest worst-case outcome. If the firm uses the maximum opportunity or minimax regret criterion and cares mainly about limiting the largest possible regret from a wrong choice, it should choose Strategy B, the US coffee focus, because this option has the smallest maximum regret. If the firm trusts the estimated probabilities for demand and cost conditions and uses the Bayes expected value rule to maximise expected annual profit, it should again choose Strategy B, since it has the highest expected monetary value.

The decision tree representation confirms the structure of the problem and helps visualise the role of the two independent events demand and costs, while the comparison and the overall evaluation clarify how each rule reflects a particular attitude to risk and uncertainty. Taking all criteria together, a coherent summary conclusion is that Strategy

B, the US coffee focus, is the most balanced choice for this firm. It avoids the potential losses of Strategy A in adverse states, performs better than Strategy C in terms of expected profit and at the same time keeps the maximum regret at a relatively low level. For a firm that wants to use probability information responsibly and avoid extreme risk while still pursuing strong expected returns, formulating the final decision as choosing Strategy B is consistent with the overall analysis of the decision-making problem.

Back to top

State of nature	Combined scenario	Probability	Strategy A Profit	Strategy B Profit	Strategy C Profit
S1	High demand, favourable costs	0.18	4700	3400	2800
S2	High demand, unfavourable costs	0.12	3500	2400	2100
S3	Moderate demand, favourable costs	0.27	2500	2600	2300
S4	Moderate demand, unfavourable costs	0.18	1300	1600	1600
S5	Weak demand, favoura-	0.15	900	2000	1900

State	Scenario	Strategy A Regret	Strategy B Regret	Strategy C Regret
S1	High demand, favourable costs	$4700 - 4700 = 0$	$4700 - 3400 = 1300$	$4700 - 2800 = 1900$
S2	High demand, unfavourable costs	$3500 - 3500 = 0$	$3500 - 2400 = 1100$	$3500 - 2100 = 1400$
S3	Moderate demand, favourable costs	$2600 - 2500 = 100$	$2600 - 2600 = 0$	$2600 - 2300 = 300$
S4	Moderate demand, unfavourable costs	$1600 - 1300 = 300$	$1600 - 1600 = 0$	$1600 - 1600 = 0$
S5	Weak demand, favourable costs	$2000 - 900 = 1100$	$2000 - 2000 = 0$	$2000 - 1900 = 100$
S6	Weak demand, unfavourable costs	$1200 - (-300) = 1500$	$1200 - 1000 = 200$	$1200 - 1200 = 0$

Capítulo 4

Example 3

4.0.1. Problem description

A Colombian agro-export firm that sells fruit-based products and coffee is planning its strategy for the next five years. Management wants to separate the way it makes decisions about revenue from the way it makes decisions about expenses. For the revenue side, the firm is considering two different market and pricing strategies. The first revenue strategy is a premium positioning aimed at high-income consumers and European specialty stores, with higher prices but a stronger dependence on favourable demand conditions. This will be called the premium revenue strategy. The second revenue strategy is a volume-based strategy focused on larger retail chains and regional markets, with more moderate prices but more stable volumes. This will be called the volume revenue strategy.

For the expense side, the firm must choose how to organise its production and logistics. One option is to invest in a more automated and contracted logistics system, which requires higher fixed costs but reduces exposure to large increases in variable costs. This will be called the stable cost strategy. The other option is to rely on flexible outsourcing and spot contracts for transport and inputs, which can be cheaper when global conditions are calm but much more expensive when there are cost shocks. This will be called the flexible cost strategy. The firm must choose one revenue strategy and one cost strategy, so it effectively chooses a combination such as premium revenue with stable costs or volume revenue with flexible costs.

The firm faces two independent sources of uncertainty. The first is global demand for its products, which can be high, moderate or weak. When demand is high, if the firm chooses the premium revenue strategy it expects annual revenue of 6000, while with the volume revenue strategy it expects annual revenue of 5500. When demand is moderate, the premium revenue strategy is expected to generate revenue of 4500 and the volume revenue strategy revenue of 4300. When demand is weak, the premium revenue strategy is expected to generate revenue of 3200 and the volume revenue strategy revenue of 3800. All revenues are measured in thousands of US dollars per year. Based on market analysis, management estimates that the probability of high demand is 0.30, the probability of moderate demand is 0.45 and the probability of weak demand is 0.25.

The second source of uncertainty is the global cost environment, which affects production and logistics expenses. The firm considers two possible cost conditions. In a favourable cost environment, with relatively low input and shipping costs, yearly expenses are expected to be 2000 if the firm chooses the stable cost strategy and 1500 if it chooses the flexible cost strategy. In an unfavourable cost environment, with high fuel prices and congested shipping routes, yearly expenses are expected to be 2600 under the stable cost strategy and 3400 under the flexible cost strategy. These expense values are also measured in thousands of US dollars per year. The firm estimates that the probability of favourable costs is 0.60 and the probability of unfavourable costs is 0.40. It assumes that global demand conditions and the global cost environment are independent.

Annual profit is defined as revenue minus expenses. Therefore, for each year, profit is determined by the combination of the demand scenario that occurs, the cost scenario that occurs and the pair of decisions chosen by the firm (one decision for revenue and another for costs). Because demand and cost conditions are independent, the probability of any combined state of nature is the product of the probability of the corresponding demand level and the probability of the corresponding cost level. Management wants to compare the different revenue–cost combinations using several decision rules, including Maximax, Maximin, the maximum opportunity (minimax regret) criterion and the Bayes expected value rule, and then decide which combination of revenue strategy and cost strategy is most coherent given its preferences about risk and its probability assessments.

4.0.2. Solution

(C2) Interpreting the parts of the decision problem

We interpret the narrative and identify the key components of the decision situation.

(C3) Building the payoff table from the revenue and expense information

From the problem description, we first organise the revenue and expense information for each separate event and decision. All amounts are in thousands of US dollars per year.

Revenue as a function of demand for each revenue strategy:

Expenses as a function of cost conditions for each cost strategy:

Element	Description
Revenue decision alternatives	Premium revenue strategy (higher prices, more sensitive to demand). Volume revenue strategy (moderate prices, more stable volumes).
Cost decision alternatives	Stable cost strategy (automation and long-term contracts, more stable expenses). Flexible cost strategy (outsourcing and spot contracts, cheaper in calm periods but risky under cost shocks).
Combined decision alternatives	A: Premium revenue + stable costs. B: Premium revenue + flexible costs. C: Volume revenue + stable costs. D: Volume revenue + flexible costs.
Independent events	Demand event with three levels high, moderate and weak, which determines revenue. Cost event with two levels favourable and unfavourable, which determines expenses.
States of nature	Each state of nature is a combination of one demand level and one cost level, for example high demand with favourable costs or weak demand with unfavourable costs.
Probabilities	Demand probabilities are 0.30 for high demand, 0.45 for moderate demand and 0.25 for weak demand. Cost probabilities are 0.60 for favourable costs and 0.40 for unfavourable costs. Demand and cost events are independent, so each combined state's probability is the product of the relevant demand probability and cost probability.
Consequences	Annual profit in thousands of US dollars, calculated as revenue minus expenses, for each combined revenue–cost decision and each combined state of nature.

Demand level	Probability	Premium revenue strategy	Volume revenue strategy
High	0.30	6000	5500
Moderate	0.45	4500	4300
Weak	0.25	3200	3800

Cost level	Probability	Stable cost strategy	Flexible cost strategy
Favourable	0.60	2000	1500
Unfavourable	0.40	2600	3400

Because demand and cost conditions are independent, a state of nature is described by a pair of events. We define six combined states: S1 is high demand with favourable costs, S2 is high demand with unfavourable costs, S3 is moderate demand with favourable costs, S4 is moderate demand with unfavourable costs, S5 is weak demand with favourable costs and S6 is weak demand with unfavourable costs. The joint probabilities are

$$P(S1) = 0{,}30 \cdot 0{,}60 = 0{,}18,$$
$$P(S2) = 0{,}30 \cdot 0{,}40 = 0{,}12,$$
$$P(S3) = 0{,}45 \cdot 0{,}60 = 0{,}27,$$
$$P(S4) = 0{,}45 \cdot 0{,}40 = 0{,}18,$$
$$P(S5) = 0{,}25 \cdot 0{,}60 = 0{,}15,$$
$$P(S6) = 0{,}25 \cdot 0{,}40 = 0{,}10.$$

Profit for each combined decision and state of nature is revenue minus expenses. We label the combined decisions as follows: A is premium revenue with stable costs, B is premium revenue with flexible costs, C is volume revenue with stable costs and D is volume revenue with flexible costs. For example, in S1 there is high demand and favourable costs. Under decision A, revenue is 6000 and expenses are 2000, so profit is $6000 - 2000 = 4000$. Under decision B, revenue is 6000 and expenses are 1500, so profit is $6000 - 1500 = 4500$. In S6 there is weak demand and unfavourable costs. Under decision B, revenue is 3200 and expenses are 3400, so profit is $3200 - 3400 = -200$.

The resulting payoff table of profit (in thousands of US dollars) is:

State	Combined scenario	Probability	A: Premium stable	B: Premium flexible	C: Volume stable	D: Volume flexible
S1	High demand, favourable costs	0.18	4000	4500	3500	4000
S2	High demand, unfavourable costs	0.12	3400	2600	2900	2100
S3	Moderate demand, favourable costs	0.27	2500	3000	2300	2800
S4	Moderate demand, unfavourable costs	0.18	1900	1100	1700	900
S5	Weak demand, favourable costs	0.15	1200	1700	1800	2300
S6	Weak demand, unfavourable costs	0.10	600	-200	1200	400

(C4) Maximax rule for decision making without probabilities

Under the Maximax rule, the decision maker focuses on the best possible profit of each combined decision and ignores probabilities and worst cases.

$$\text{máx}\{4000,\ 3400,\ 2500,\ 1900,\ 1200,\ 600\} = 4000.$$

$$\text{máx}\{4500,\ 2600,\ 3000,\ 1100,\ 1700,\ -200\} = 4500.$$

$$\text{máx}\{3500,\ 2900,\ 2300,\ 1700,\ 1800,\ 1200\} = 3500.$$

$$\text{máx}\{4000,\ 2100,\ 2800,\ 900,\ 2300,\ 400\} = 4000.$$

The largest of these best profits is 4500, which corresponds to decision B in state S1. According to the Maximax rule, the firm should choose *decision B (premium revenue with flexible costs).*

(C5) Maximin rule for decision making without probabilities

Under the Maximin rule, the decision maker looks at the worst profit of each combined decision and then chooses the decision whose worst-case outcome is the strongest.

$$\text{mín}\{4000,\ 3400,\ 2500,\ 1900,\ 1200,\ 600\} = 600.$$

$$\text{mín}\{4500,\ 2600,\ 3000,\ 1100,\ 1700,\ -200\} = -200.$$

$$\text{mín}\{3500,\ 2900,\ 2300,\ 1700,\ 1800,\ 1200\} = 1200.$$

$$\text{mín}\{4000,\ 2100,\ 2800,\ 900,\ 2300,\ 400\} = 400.$$

We compare these worst-case values and choose the highest one,

$$\text{máx}\{600,\ -200,\ 1200,\ 400\} = 1200.$$

Under the Maximin rule, the firm should choose *decision C (volume revenue with stable costs)*, because it offers the strongest worst-case profit.

(C6) Maximum opportunity criterion (minimax regret)

The maximum opportunity or minimax regret criterion focuses on regret, defined as the loss from not having chosen the best decision in a given state of nature. For each state, we identify the best profit and then compute regret as the best profit in that state minus the profit of the chosen decision.

$$\begin{aligned}
\text{S1}&: \text{máx}\{4000,\ 4500,\ 3500,\ 4000\} = 4500,\\
\text{S2}&: \text{máx}\{3400,\ 2600,\ 2900,\ 2100\} = 3400,\\
\text{S3}&: \text{máx}\{2500,\ 3000,\ 2300,\ 2800\} = 3000,\\
\text{S4}&: \text{máx}\{1900,\ 1100,\ 1700,\ 900\} = 1900,\\
\text{S5}&: \text{máx}\{1200,\ 1700,\ 1800,\ 2300\} = 2300,\\
\text{S6}&: \text{máx}\{600,\ -200,\ 1200,\ 400\} = 1200.
\end{aligned}$$

The regret table in thousands of US dollars is:

For each decision, we identify its maximum regret across all states:

$$\begin{aligned}
\text{máx}\,\text{Regret}(A) &= \text{máx}\{500,\ 0,\ 500,\ 0,\ 1100,\ 600\} = 1100,\\
\text{máx}\,\text{Regret}(B) &= \text{máx}\{0,\ 800,\ 0,\ 800,\ 600,\ 1400\} = 1400,\\
\text{máx}\,\text{Regret}(C) &= \text{máx}\{1000,\ 500,\ 700,\ 200,\ 500,\ 0\} = 1000,\\
\text{máx}\,\text{Regret}(D) &= \text{máx}\{500,\ 1300,\ 200,\ 1000,\ 0,\ 800\} = 1300.
\end{aligned}$$

The minimax regret rule chooses the decision with the smallest of these maximum regrets, which is decision C. The maximum opportunity criterion therefore recommends *decision C (volume revenue with stable costs).*

(C7) Using probabilities and expected value (Bayes decision rule)

The Bayes decision rule with expected monetary value (EMV) multiplies each profit by the probability of its state and adds the results. We use the joint probabilities of S1 to S6.

$$\begin{aligned}
EMV_A &= 0{,}18(4000) + 0{,}12(3400) + 0{,}27(2500) + 0{,}18(1900) + 0{,}15(1200) + 0{,}10(600)\\
&= 720 + 408 + 675 + 342 + 180 + 60 = 2385.
\end{aligned}$$

$$\begin{aligned}
EMV_B &= 0{,}18(4500) + 0{,}12(2600) + 0{,}27(3000) + 0{,}18(1100) + 0{,}15(1700) + 0{,}10(-200)\\
&= 810 + 312 + 810 + 198 + 255 - 20 = 2365.
\end{aligned}$$

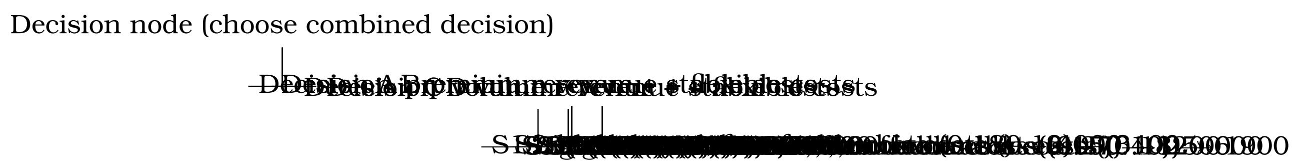
$$\begin{aligned}
EMV_C &= 0{,}18(3500) + 0{,}12(2900) + 0{,}27(2300) + 0{,}18(1700) + 0{,}15(1800) + 0{,}10(1200)\\
&= 630 + 348 + 621 + 306 + 270 + 120 = 2295.
\end{aligned}$$

$$\begin{aligned}
EMV_D &= 0{,}18(4000) + 0{,}12(2100) + 0{,}27(2800) + 0{,}18(900) + 0{,}15(2300) + 0{,}10(400)\\
&= 720 + 252 + 756 + 162 + 345 + 40 = 2275.
\end{aligned}$$

These values are expected annual profits in thousands of US dollars. Since $EMV_A = 2385$ is the highest expected value, the Bayes EMV rule recommends *decision A (premium revenue with stable costs).*

(C9) Tree diagram of the decision-making problem

The same decision situation can be represented as a decision tree. The firm first chooses a combined decision (revenue strategy and cost strategy), then the demand level is realised and then the cost level is realised. Because demand and costs are independent, their probabilities multiply along each path. For simplicity, we show the combined states with their joint probabilities and profits.



This diagram makes explicit the sequence of decisions and uncertain events, as well as the joint probabilities and profits associated with each possible path.

(C8) Comparing Bayes, Maximin, Maximax and minimax regret

We summarise the recommendations from the different decision rules:

- Maximax, which is very optimistic and focuses on the best possible profit, recommends decision B (premium revenue + flexible costs) because it can reach 4500.

- Maximin, which is very cautious and focuses on the worst-case profit, recommends decision C (volume revenue + stable costs) because its worst case is 1200.

- The maximum opportunity or minimax regret rule, which focuses on limiting the largest possible regret, recommends decision C because its maximum regret is 1000, smaller than the maximum regrets of A, B and D.

- The Bayes EMV rule, which uses the full joint distribution of probabilities and profits, recommends decision A (premium revenue + stable costs) because it has the highest expected monetary value, 2385.

The comparison shows that rules ignoring probabilities and focusing on extreme outcomes can produce different recommendations from rules that use all probability information. Decision B is favoured by a very optimistic view, decision C by a very cautious view and by regret minimisation, while decision A is favoured by the expected value criterion.

(C10) Evaluating all possibilities in the decision problem

Looking across all rules, the firm can evaluate each combined decision in terms of risk and reward. Decision A, premium revenue with stable costs, offers strong profits in favourable states without negative outcomes and achieves the highest expected profit, but it does not have the strongest worst-case profit. Decision B, premium revenue with flexible costs, has the highest possible profit when demand is high and costs are favourable but exposes the firm to a negative profit of -200 when demand is weak and costs are unfavourable, and therefore has a high maximum regret. Decision C, volume revenue with stable costs, never performs extremely badly or extremely well. It offers the strongest worst-case profit and the smallest maximum regret, but its expected value is somewhat lower than that of decision A. Decision D, volume revenue with flexible costs, lies between the others in many states but does not dominate any of them in terms of either expected value, worst case or regret. This evaluation highlights how attitudes towards risk, regret and the role of probabilities lead to different recommended actions, and how separating revenue and expense decisions creates a richer set of trade-offs.

(C1) Formulating the final decision and summarising all criteria

Based on each of the decision criteria used in the solution, the recommended combined decision would be as follows. If the firm decides according to the Maximax rule and adopts a highly optimistic and risk-seeking attitude, it should choose decision B, premium revenue with flexible costs, because this combination can reach the highest profit among all combined states. If the firm decides according to the Maximin rule and wants to protect itself against the worst possible profit, it should choose decision C, volume revenue with stable costs, because this combination has the strongest worst-case outcome. If the firm uses the maximum opportunity or minimax regret criterion and cares mainly about limiting the largest possible regret from a wrong choice, it should again choose decision C, since its maximum regret is the smallest among all options. If the firm trusts the estimated probabilities for demand and cost conditions and uses the Bayes expected value rule to maximise expected annual profit, it should choose decision A, premium revenue with stable costs, because this combination has the highest expected monetary value.

The decision tree representation confirms how the two independent events, demand and costs, interact with the two separate decisions on revenue and expenses. The comparison and evaluation clarify how each decision rule reflects a particular attitude toward risk and uncertainty. Taking all criteria together, a coherent summary conclusion is that decision A, premium revenue with stable costs, is the most attractive option for a firm that wants to maximise expected profit while avoiding extreme downside risk, whereas decision C, volume revenue with stable costs, is more appropriate for a firm that prioritises robustness of the worst-case outcome and regret minimisation. For a firm that wishes to use probability information responsibly but still give weight to security, stating its final decision as choosing premium revenue with stable costs and clearly understanding how this differs from the more conservative choice volume revenue with stable costs is consistent with the overall analysis of this two-decision problem.

Back to top

State	Scenario	A Regret	B Regret	C Regret	D Regret
S1	High demand, favourable costs	$4500 - 4000 = 500$	$4500 - 4500 = 0$	$4500 - 3500 = 1000$	$4500 - 4000 = 500$
S2	High demand, unfavourable costs	$3400 - 3400 = 0$	$3400 - 2600 = 800$	$3400 - 2900 = 500$	$3400 - 2100 = 1300$
S3	Moderate demand, favourable costs	$3000 - 2500 = 500$	$3000 - 3000 = 0$	$3000 - 2300 = 700$	$3000 - 2800 = 200$
S4	Moderate demand, unfavourable costs	$1900 - 1900 = 0$	$1900 - 1100 = 800$	$1900 - 1700 = 200$	$1900 - 900 = 1000$
S5	Weak demand, favourable costs	$2300 - 1200 = 1100$	$2300 - 1700 = 600$	$2300 - 1800 = 500$	$2300 - 2300 = 0$
S6	Weak	$1200 - 600 = 600$	$1200 - (-200) = 1400$	$1200 - 1200 = 0$	$1200 - 400 = 800$